

THE 20 DOLLAR GREASE FIRE BLANKET: EVERY NUMBER

A 20 dollar fiberglass blanket smothers a stovetop grease fire that a single cup of water turns into a fireball. What the blanket is, how you use it, and why a cover beats water.

WHAT IT IS

A kitchen fire blanket is a 40 inch square of woven glass fiber in a pouch with two pull tabs. Glass fiber does not burn, and a properly woven sheet does not let air through. Lay it over a burning pan and it cuts off the oxygen, so the flame dies in a few seconds.

About 20 dollars buys a good one. Prepared Hero sells one at Target for about 26 dollars. Store brands and multipacks bring it under 15 dollars.

HOW TO USE IT

- 1** Mount the pouch where you can grab it without reaching past the stove, ideally on the way out of the kitchen. Do not hang it above the range hood.
- 2** Pull both tabs straight down. The blanket drops out of the pouch.
- 3** Hold it by the top corners. Fold the top edge back over your hands so the fabric is between your knuckles and the fire.
- 4** Hold it up in front of you like a shield.
- 5** Lay it over the pan starting from the edge nearest you, then let it settle over the far side. Never throw it.
- 6** Turn the burner off if the knob is reachable without leaning over the pan. If the knob is behind the fire, leave it.
- 7** Step back. Leave the pan alone. Do not carry a burning pan to the sink or the door.

WHY A COVER WORKS

The liquid oil is not what burns. The flame sits above the surface, where hot oil vapor mixes with air. Cut the air off and the flame goes out for lack of oxygen.

LEAVE THE COVER ON UNTIL THE PAN IS COLD

A pan of oil holds a lot of heat. The oil underneath can stay far above its flash point, so lifting the cover to peek lets air reach vapor that is still hot enough to burn, and the fire relights. Oil that has already burned relights more easily than fresh oil.

- ☐ **The cover stays on until the pan is completely cool.** That is the wording the National Fire Protection Association uses.
- ☐ For a pot full of oil, that can take well over an hour.
- ☐ Do not peek, and do not touch the pan to check.
- ☐ Turning the burner off matters too. It keeps pumping heat into the oil after the flame is gone.

A LID OR A TRAY DOES THE SAME JOB

On a pan that has a lid, slide the lid over and turn off the burner. No lid handy? A metal baking sheet works the same way. The blanket earns its place because it covers what a lid cannot, like a wide skillet or a spill burning on the cooktop. It has to cover the whole fire. Never throw flour on it, because a cloud of flour can flash.

HOW OIL CATCHES FIRE ON ITS OWN

A grease fire starts without a match. The National Institute of Standards and Technology put corn oil and peanut oil in real cookware on an electric range, left the burner on high, and measured it.

about 18 min	Time for a small pan with less than a cup of oil to catch fire. Nothing touched it.
about 80 min	Time for a big pot holding about 4 liters. The oil just kept climbing past its ignition point.
254 C	Corn oil flash point. Above that, vapor off the surface can catch from a spark or burner flame.
392 C	Corn oil autoignition temperature. At that point the vapor lights with no spark at all.
350 C	Pan bottom in one set of tests when the first flame appeared. That is over 660 degrees Fahrenheit.

THE WARNING SIGN IS SMOKE

A University of Maryland fire protection study cites a Canadian test where smoke started near 220 degrees Celsius, well before ignition. Smoke rising off a pan of oil means it is well past cooking temperature and on its way. The United States Fire Administration rule is to turn the burner off the moment you see smoke or the oil starts to boil.

4x wide	How far the flame grew past the width of the pan as the oil kept heating.
10x tall	At the peak, flames stood more than 10 times as tall as the pan was wide.
about 1 min	For the small pan, time from the first flame until the oil foamed over the rim.
20,000 W	Heat measured about two and a half feet above the pan, per square meter, roughly the level where particle board can ignite. So a pan fire can light the cabinets.

WHAT WATER DOES TO BURNING OIL

Water is denser than oil. The corn oil in those tests weighed about 92 percent as much as water, volume for volume. So water poured onto a burning pan does not sit on top and cool the flame. It sinks straight through the oil to the bottom of the pan, which is sitting near 350 degrees Celsius.

1,700x	How much the volume grows when liquid water flashes to steam at normal pressure.
15 mL	One tablespoon of water. Flash it to steam and you get roughly 25 liters of gas, expanding from under the oil in a fraction of a second.
240 mL	A full cup of water. If all of it flashed, that is about 400 liters of steam. It does not need to. The first splash does the damage.

The steam blows the oil out of the pan as a spray of fine droplets. Each droplet is already hotter than the oil flash point, and now each one has air on all sides. The spray lights as one sheet. A pan fire becomes a ball of flame that fills the space over the stove and reaches the ceiling. The person holding the cup is standing at arm's length from that, and the face and hands take the flash.

NEVER PUT THESE ON A GREASE FIRE

- Water, even a splash.
- A wet towel that is dripping.
- A glass of anything mostly water. Milk and juice count.
- Flour or any powder that can flash into a cloud.

In a public demonstration in Mankato, Minnesota, in 2023, a single cup of water went into a burning pan and the fire spread across the entire mock kitchen almost instantly.

THE SCALE OF THE PROBLEM

The National Fire Protection Association counts these fires. From 2017 through 2021, in the United States:

158,000	Reported home fires a year caused by cooking, on average. That is 44 percent of all home fires.
470 / yr	Deaths a year on average. Plus more than 4,000 injured and over a billion dollars in damage.
53% / 88%	Share of those fires involving ranges and cooktops, and share of the deaths. Electric ranges had a higher rate than gas.
1,470	Thanksgiving fires, the worst day of the year by a wide margin, against a daily average of 430.
4 out of 5	Fires that never got past the pan they started in. Most were small.

TWO NUMBERS THAT AIM YOU AT THE COVER

- Among cooking fires that started with food or grease, oil and grease were the first thing to catch in about half.
- More than half of the people injured in cooking fires were hurt while trying to fight the fire themselves.

WHERE BLANKETS HAVE FAILED

A blanket is not magic. A government test and a broadcast test both found blankets that failed.

16 of 22	Blankets that caught fire themselves when the Dutch consumer product safety authority tested them on pan fires in 2014.
the other 6	Relit when the blanket was lifted after 17 minutes. The authority's advice afterward was to use a lid on an oil fire.
2 of 6	Blankets bought online that flames came through, when the West Midlands fire service tested them for a British television investigation.

A blanket burns when its coating burns, or when the weave is loose enough for vapor to pass through and light on top. A fire relights when the blanket comes off too early, the same mistake as lifting a lid. Seventeen minutes was not long enough for a pan holding liters of oil.

THE BUYING RULE

- ☐ Buy from a maker who names the standard and the test lab, and read its instructions.
- ☐ Skip the cheapest unbranded listing.
- ☐ The European standard for these blankets is number eighteen sixty nine. To pass, a blanket must put out a fire in 3 liters of cooking oil, stay intact, and not catch fire itself. It asks for each side to reach at least 1 meter, about 39 inches, so size alone will not prove it complies.
- ☐ The 2019 version treats the blanket as single use. Once it has been on a fire and fully cooled, it goes in the trash and you buy another.
- ☐ One maker claims the fabric handles heat up to 580 degrees Celsius. That is the maker's claim, not an independent test.

GET OUT AND CALL NINE ONE ONE

The line where you stop fighting comes earlier than most people think. Any one of these signs, and you leave:

- Flames taller than the pan is wide.
- Fire on the cabinets or the wall.
- Smoke gathering at the ceiling.
- Flame coming through the blanket. If you see that, you are done fighting.

Get everyone out, close the kitchen door behind you, and call nine one one from outside. The closed door slows the fire and the smoke.

THE EXTINGUISHER AND ITS LIMITS

A basic home extinguisher is rated for wood, liquids, and electrical fires. Keep one, and know its limits. The dry powder comes out under pressure, and aimed into a pan from close range that jet can throw burning oil over the rim, which is the water problem by another route. British fire services tell people not to use an extinguisher on a pan of oil at all.

- ☐ Stand 6 to 8 feet back, and keep your exit behind you.
- ☐ Only fight a fire that is small and not spreading, after the fire department has been called.
- ☐ So the cover is for the pan, and nothing gets sprayed into it.

IF A PERSON CATCHES FIRE

The blanket covers a person too. If a sleeve catches, the person drops to the floor and rolls, and you wrap the blanket over them from the shoulders down. The fabric between the flame and the air does the same job it does on the pan.

WHAT A BAD FIRE COSTS

88,000	Average homeowner fire claim in the United States from 2019 through 2023, in dollars, per the Insurance Information Institute.
under 40	Dollars for a blanket and a lid together.
1 in 4+	More than a quarter of the people killed in cooking fires were asleep.
18 min	How fast oil lit itself in the government tests, with nobody touching it. The cheapest fix of all is to never leave a pan of oil on a burner.

Smoke reaches a sleeping person before any flame does, and that has its own physics, worked out in the video about why most fire deaths are not from flames.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

Subscribe so the next kitchen fire does not catch you reaching for the wrong thing.

Sketchy Survival 101. Physics first survival. Every number on this sheet comes straight from the video, with its hedges kept: dated tests, single maker claims, and typical values that are not safety limits. This sheet is for education. In a real fire, if it is spreading, get out and call nine one one.